

is the most abundant natural monosaccharide because it is less glycosylated with proteins than other monosaccharides.^{[29][30]} Another hypothesis is that glucose, being the only D-aldohexose that has all five hydroxy substituents in the equatorial position in the form of β-D-glucose, is more readily accessible to chemical reactions,^[31] for example, for esterification^[32] or acetal formation.^[33] For this reason, D-glucose is also a highly preferred building block in natural polysaccharides (glycans). Polysaccharides that are composed solely of Glucose are termed glucans.

Glucose is produced by plants through the photosynthesis using sunlight, water and carbon dioxide and can be used by all living organisms as an energy and carbon source. However, most glucose does not occur in its free form, but in the form of its polymers, i.e. lactose, sucrose, starch and others which are energy reserve substances, and cellulose and chitin, which are components of the cell wall in plants or fungi and arthropods, respectively. These polymers are degraded to glucose during food intake by animals, fungi and bacteria using enzymes. All animals are also able to produce glucose themselves from certain precursors as the need arises. Nerve cells, cells of the renal medulla and erythrocytes depend on glucose for their energy production.^[34] In adult humans, there are about 18 g of glucose,^[35] of which about 4 g are present in the blood.^[36] Approximately 180 to 220 g of glucose are produced in the liver of an adult in 24 hours.^[35]

Many of the long-term complications of diabetes (e.g., blindness, kidney failure, and peripheral neuropathy) are probably due to the glycation of proteins or lipids.^[37] In contrast, enzyme-regulated addition of sugars to protein is called glycosylation and is essential for the function of many proteins.^[38]

Uptake

Ingested glucose initially binds to the receptor for sweet taste on the tongue in humans. This complex of the proteins T1R2 and T1R3 makes it possible to identify glucose-containing food sources. Glucose mainly comes from food - about 300 g per day are produced by conversion of food,^[39] but it is also synthesized from other metabolites in the body's cells. In humans, the breakdown of glucose-containing polysaccharides happens in part already during chewing by means of amylase, which is contained in saliva, as well as by maltase, lactase and sucrase on the brush border of the small intestine. Glucose is a building block of many carbohydrates and can be split off from them using certain enzymes. Glucosidases, a subgroup of the glycosidases, first catalyze the hydrolysis of long-chain glucose-containing polysaccharides, removing terminal glucose. In turn, disaccharides are mostly degraded by specific glycosidases to glucose. The names of the degrading enzymes are often derived from the particular poly- and disaccharide; inter alia, for the degradation of polysaccharide chains there are amylases (named after amylose, a component of starch), cellulases (named after cellulose), chitinases (named after chitin) and more. Furthermore, for the cleavage of disaccharides, there are maltase, lactase, sucrase, trehalase and others. In humans, about 70 genes are known that code for glycosidases. They have functions in the digestion and degradation of glycogen, sphingolipids, mucopolysaccharides and poly(ADP-ribose). Humans do not produce cellulases, chitinases and trehalases, but the bacteria in the gut flora do.

In order to get into or out of cell membranes of cells and membranes of cell compartments, glucose requires special transport proteins from the major facilitator superfamily. In the small intestine (more precisely, in the jejunum),^[40] glucose is taken up into the intestinal epithelial cells with the help of glucose transporters^[41] via a secondary active transport mechanism called sodium ion-glucose symport via the sodium/glucose cotransporter 1.^[42] The further transfer occurs on the basolateral side of the intestinal epithelial cells via the glucose transporter GLUT2,^[42] as well as their uptake into liver cells, kidney cells, cells of the islets of Langerhans, nerve cells, astrocytes and tanyocytes.^[43] Glucose enters the liver via the vena portae and is stored there as a cellular glycogen.^[44] In the liver cell, it is phosphorylated by glucokinase at position 6 to glucose-6-phosphate, which can not leave the cell. With the help of glucose-6-phosphatase, glucose-6-phosphate is converted back into glucose exclusively in the liver, if necessary, so that it is available for maintaining a sufficient blood glucose concentration. In other cells, uptake happens by passive transport through one of the 14 GLUT proteins.^[42] In the other cell types, phosphorylation occurs through a hexokinase, whereupon glucose can no longer diffuse out of the cell.

The glucose transporter GLUT1 is produced by most cell types and is of particular importance for nerve cells and pancreatic β-cells.^[42] GLUT3 is highly expressed in nerve cells.^[42] Glucose from the bloodstream is taken up by GLUT4 from muscle cells (of the skeletal muscle^[45] and heart muscle) and fat cells.^[46] GLUT14 is formed exclusively in testes.^[42] Excess glucose is broken down and converted into fatty acids, which are stored as triacylglycerides. In the kidneys, glucose in the urine is absorbed via SGLT1 and SGLT2 in the apical cell membranes and transmitted via GLUT2 in the basolateral cell membranes.^[47] About 90% of kidney glucose reabsorption is via SGLT2 and about 3% via SGLT1.^[48]

Biosynthesis

In plants and some prokaryotes, glucose is a product of photosynthesis.^[49] Glucose is also formed by the breakdown of polymeric forms of glucose like glycogen (in animals and mushrooms) or starch (in plants). The cleavage of glycogen is termed glycogenolysis, the cleavage of starch is called starch degradation.^[50]

The metabolic pathway that begins with molecules containing two to four carbon atoms (C) and ends in the glucose molecule containing six carbon atoms is called gluconeogenesis and occurs in all living organisms. The smaller starting materials are the result of other metabolic pathways. Ultimately almost all biomolecules come from the assimilation of carbon dioxide in plants during photosynthesis.^[51] The free energy of formation of α-D-glucose is 917.2 kilojoules per mole.^[52] In humans, gluconeogenesis occurs in the liver and kidney,^[53] but also in other cell types. In the liver about 150 g of glycogen are stored, in skeletal muscle about 250 g.^[54] However, the glucose released in muscle cells upon cleavage of the glycogen can not be delivered to the circulation because glucose is phosphorylated by the hexokinase, and a glucose-6-phosphatase is not expressed to remove the phosphate group. Unlike for glucose, there is no transport protein for glucose-6-phosphate. Gluconeogenesis allows the organism to build up glucose from other metabolites, including lactate or certain amino acids, while consuming energy. The renal tubular cells can also produce glucose.

Glucose degradation

In humans, glucose is metabolised by glycolysis^[55] and the pentose phosphate pathway.^[56] Glycolysis is used by all living organisms,^{[57][58]} with small variations, and all organisms generate energy from the breakdown of monosaccharides.^[57] In the further course of the metabolism, it can be completely degraded via oxidative decarboxylation, the Krebs cycle (synonym *citric acid cycle*) and the respiratory chain to water and carbon dioxide. If there is not enough oxygen available for this, the glucose degradation in animals occurs anaerobic to lactate via lactic acid fermentation and releases less energy. Muscular lactate enters the liver through the bloodstream in mammals, where gluconeogenesis occurs (Cori cycle). With a high supply of glucose, the metabolite acetyl-CoA from the Krebs cycle can also be used for fatty acid synthesis.^[59] Glucose is also used to replenish the body's glycogen stores, which are mainly found in liver and skeletal muscle. These processes are hormonally regulated.